

"AI-Powered Idea Evolution Platform"

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1. Executive Summary

IdeaDNA is an AI-powered Idea Evolution Platform that transforms how developers, founders, and students generate, evolve, and implement innovative project ideas. Unlike existing AI tools that only generate ideas, IdeaDNA evolves them through structured mutation, reverse-engineers developer thinking from code, and produces production-ready blueprints with live UI mockups.

The platform operates in three primary modes: Idea Generator (problem-to-solution), Evolution Engine (idea-to-upgraded-versions), and Code Archaeology (code-to-thinking-process). Each mode leverages a multi-agent LLM system built on LangChain, with lazy-loading architecture optimizing costs by 80%.

Key Innovation: A curated database of 500+ platform-specific ideas combined with AI-powered real-time personalization ensures every output is tailored to the user's skill level, budget, team size, and timeline.

2. Problem Statement & Market Analysis

2.1 Core Problems

- Ideation Gap: Users struggle to find unique, non-obvious project ideas that stand out from competitors
- Evolution Stagnation: Existing AI tools generate ideas but fail to evolve them into multiple optimized versions
- Thinking Opacity: No way to understand how experienced developers architect solutions from existing code
- Blueprint Vacuum: Generated ideas lack actionable implementation plans, tech stacks, and database schemas
- Resource Waste: Generating heavy assets for all ideas when users only select one
- Personalization Deficit: Generic outputs that ignore user's skill level, budget, team size, and timeline

2.2 Market Analysis

- TAM (Total Addressable Market): 26M+ developers worldwide, 10M+ students, 5M+ startup founders
- SAM (Serviceable Addressable Market): 5M active hackathon participants, 2M indie hackers, 1M CS students annually
- SOM (Serviceable Obtainable Market): 100K active users in Year 1 target
- Competitors: ChatGPT (generic), GitHub Copilot (code-only), Notion AI (docs-only), Figma AI (design-only)
- Competitive Advantage: Only platform with idea evolution + code archaeology + blueprint generation in one pipeline

3. Vision & Mission

Vision

To become the world's first AI-powered innovation platform that doesn't just generate ideas — it evolves them, archives them, and teaches users how to think like world-class developers through reverse-engineering and structured evolution.

Mission

Empower every developer, founder, and student to go from 'I have a problem' to 'I have a production-ready blueprint' in under 10 minutes, while learning the thinking patterns behind great software architecture.

Tagline

'Don't just generate ideas. Evolve them.'

4. Target Users & Personas

Hackathon Harry (PRIMARY)

College student, 20, Intermediate skills, \$0 budget, 3-person team, 24-48h timeline. Needs: Unique project that impresses judges. Wants: Quick MVP ideas, visual wow factor.

Startup Sarah (PRIMARY)

Aspiring founder, 26, Advanced skills, \$5000 budget, 2-person team, 3-month timeline. Needs: Validate multiple solution approaches. Wants: Scalable architecture, investor-ready blueprints.

Learner Leo (PRIMARY)

Self-taught developer, 22, Beginner, \$0 budget, Solo, 2-week timeline. Needs: Learn from real projects. Wants: Code archaeology, architecture explanations.

Product Manager Priya (SECONDARY)

PM at tech company, 30, Advanced, Company-funded, 5-10 team, Flexible timeline. Needs: Quick PRDs and tech recommendations. Wants: Stakeholder presentations, documentation.

Innovation Team (SECONDARY)

Corporate R&D, Mixed skills, \$50K+ budget, 10+ team, 6-month timeline. Needs: Systematic innovation pipeline. Wants: Team collaboration, idea marketplace.

5. Core Features

Feature 1: ■ Idea Generator

Input: Problem description or theme. Output: 5-8 ranked idea cards with innovation/feasibility scores. Unique: Curated database (500+ ideas per platform) + AI upgrade based on user profile. Lazy-loaded: only selected idea gets deep-dive.

Feature 2: ■ Evolution Engine

Input: Existing idea. Output: 4-6 evolved versions with mutation tags. Unique: Critic-first approach — weakness identify, then mutate. Visual: Interactive DNA Evolution Tree (React Flow). Recursive: infinite evolution depth.

Feature 3: ■ Code Archaeology

Input: Code snippet, GitHub URL, or live website. Output: Complete developer thinking analysis. Unique: 6-agent pipeline reverse-engineering psychology, architecture decisions, and evolution timeline. No competitor offers this.

Feature 4: ■ Blueprint Generator

Trigger: User selects an idea. Output: 7-section production blueprint — PRD, Tech Stack, Database Schema, API Design, User Flow, Implementation Plan. Personalized: Same idea, different user = different blueprint. Export: Markdown, PDF, OpenAPI.

Feature 5: ■ UI Mockup Generator

Trigger: User selects an idea. Output: Complete HTML/CSS mockup with Tailwind CSS. Features: Device toggle (Desktop/Tablet/Mobile), responsive preview in iframe, export as HTML/ZIP. Style variants: Modern, Minimal, Cyberpunk, Corporate, Playful.

Feature 6: ■ Innovation Scoring Engine

Cross-cutting feature. Evaluates every idea on 4 dimensions: Originality (30%), Feasibility (25%), Market Potential (25%), Complexity Match (20%). Visual: D3.js radar chart. Personalized: scores adapt to user profile.

Feature 7: ■ User Profiles & Personalization

Onboarding: 6-question survey (skill, budget, team, timeline, tech prefs, industry). Impact: Injected into every AI agent's context. Result: Same problem generates completely different outputs for different users.

Feature 8: ■ Project Persistence

Saves all projects with full history. Features: Vector search for similar ideas, Evolution lineage tracking (family tree), Export entire project as ZIP, Shareable links, Dashboard with stats and recommendations.

6. User Flow & Journey Maps

Primary Journey: Idea Generator Mode

- Step 1 — Landing: User sees 3 mode cards (Generate, Evolve, Analyze)
- Step 2 — Mode Select: User clicks 'Generate Ideas'
- Step 3 — Input: User describes problem + selects platform filter + confirms profile
- Step 4 — Processing: RetrieverAgent searches curated DB + UpgradeAgent personalizes + ScorerAgent ranks
- Step 5 — Results: 5-8 idea cards displayed with scores, tags, difficulty
- Step 6 — Selection: User clicks ONE card → status changes to SELECTED
- Step 7 — Deep Dive: Blueprint + UI Mockup tabs unlock (lazy-loaded generation starts)
- Step 8 — Review: User explores blueprint sections and UI preview
- Step 9 — Action: Export, Evolve Further, Save, or Start New

Secondary Journey: Evolution Mode

- Step 1 — Mode Select: User clicks 'Evolve Idea'
- Step 2 — Input: User describes existing idea
- Step 3 — Critic: CriticAgent analyzes weaknesses (visible to user)
- Step 4 — Evolution: EvolverAgent generates 4-6 mutated versions
- Step 5 — Tree View: React Flow interactive tree with scores
- Step 6 — Selection: User clicks ONE version
- Step 7+: Same deep-dive flow as Generator mode

7. Success Metrics & KPIs

Metric	Target (Y1)	Measurement Method
Monthly Active Users (MAU)	50,000	Analytics
Ideas Generated / Month	100,000	Database count
Selection Rate	25%	Selected / Generated
Evolution Rate	40%	Evolved / Total
Blueprint Export Rate	15%	Exports / Selected
User Retention (7-day)	30%	Return users / New users
Avg Session Duration	8 minutes	Analytics
NPS Score	50+	User survey
API Cost per Session	<\$0.50	OpenAI usage tracking
Revenue (if monetized)	\$10K MRR	Stripe dashboard

8. Non-Functional Requirements

Performance

Initial idea generation < 3 seconds. Deep-dive generation < 15 seconds. Page load < 2 seconds. 99.9% uptime.

Scalability

Handle 10,000 concurrent users. Horizontal scaling via Vercel Edge + Railway. Database connection pooling.

Security

OAuth 2.0 + JWT. AES-256 encryption for PII. API keys server-side only. Input sanitization via Zod. Rate limiting.

Accessibility

WCAG 2.1 AA compliance. Screen reader support. Keyboard navigation. Color contrast ratios > 4.5:1.

Compatibility

Chrome 90+, Firefox 88+, Safari 14+, Edge 90+. Mobile responsive: 320px+.

Cost Optimization

Lazy-loading reduces API costs by 80%. Redis caching for frequent queries. Vector search for idea deduplication.

9. Future Roadmap

Phase 2 (Months 2-3)

Team Matching System, Real-time Collaboration, Chrome Extension, Idea Marketplace, API for third-party integrations

Phase 3 (Months 4-6)

Investor Readiness Score, Startup Validation Module, Market Research Generator, Competitive Analysis Agent, Pitch Deck Auto-Builder

Phase 4 (Months 7-12)

AI Co-Founder Mode, Multi-language Support, Enterprise API, White-label Solution, Mobile App (React Native), AI Image Generation for Mockups

10. Appendices

A. Glossary

- LLM: Large Language Model (GPT-4, Claude, etc.)
- LangChain: Framework for building LLM applications with chains and agents
- Vector Search: Semantic similarity search using embeddings
- Lazy Loading: Generating heavy assets only when needed
- Multi-Agent: Multiple AI agents working in sequence or parallel
- SequentialChain: LangChain component where output of one chain feeds into next

B. Tech Stack Summary

Layer	Technology	Purpose
Frontend	Next.js 14 + TypeScript + Tailwind	App, UI, SSR
State	Zustand + React Query	Client + Server state
Backend	Next.js API Routes + tRPC	API, type safety
Database	PostgreSQL + Prisma	Relational data
Cache	Redis	Sessions, caching
AI	LangChain + OpenAI GPT-4	Agent orchestration
Vector DB	Pinecone / pgvector	Semantic search
Auth	Clerk / NextAuth	Authentication
Visuals	React Flow + D3.js + Framer Motion	Graphs, charts, animations
Deploy	Vercel + Railway	Hosting, DB

— End of PRD —